A Student Research Workflow and Collaboration Tracking Platform

Student Research Proposal • Mahbub Alam Abid • Software Engineering • Web Development • Research Management

Proposed by	Mahbub Alam Abid
Project type	Student-led research / prototype study
Estimated duration	8–10 weeks

Project Overview

I want to build a lightweight web platform that helps student researchers organize research tasks, collaboration activities, deadlines, and progress in one place. Instead of focusing on the research topic itself, the project will study how a simple workflow can reduce scattered communication and make small research teams easier to coordinate.

Student perspective
I want to keep the first version realistic for a student team: a focused research question, a reproducible experiment, and a prototype that can be discussed with a researcher and improved through collaboration.

Research Problem

Student research teams often manage papers, tasks, meeting notes, deadlines, and responsibilities across separate tools. This can make it difficult to see who is working on what and whether an important task has been delayed. I want to explore whether a focused workflow can make research coordination clearer without becoming another complicated project-management system.

Research Objectives

- Design a simple research-workflow model for small student teams.
- Develop a web prototype for projects, tasks, milestones, and collaboration notes.
- Add role-based visibility so team members can see relevant responsibilities.
- Track basic progress and overdue tasks without excessive features.
- Evaluate the prototype through realistic student research scenarios.

Research Questions

RQ1. Which small set of workflow features is most useful for student research teams?

RQ2. Does a research-specific task structure make collaboration easier than a generic task list?

RQ3. How can progress information remain useful without creating extra administrative work?

RQ4. Which interface choices help students understand project status quickly?

Proposed Methodology

Step 1. Review lightweight project-management and research-workflow concepts.

Step 2. Define entities such as projects, tasks, milestones, members, and notes.

Step 3. Design a relational database and REST-style backend.

Step 4. Build a responsive web interface and connect it to the backend.

Step 5. Test common workflows such as assigning tasks, updating status, and reviewing deadlines.

Step 6. Conduct a small usability evaluation with realistic scenarios and document improvements.

Data / Materials / Experimental Resources

Laptop/local development environment, sample research projects, task scenarios, relational database, and a small usability-test questionnaire.

Evaluation Plan

Evaluate task-completion time, number of navigation steps, usability observations, and whether users can correctly identify project status and responsibilities.

Expected Challenges

- Keeping the interface simple while supporting useful workflows.
- Avoiding unnecessary project-management features.
- Designing permissions that are understandable to students.
- Getting meaningful usability feedback from a small participant group.

Reproducibility and Documentation

I will keep the main assumptions, dataset versions, preprocessing decisions, experiment settings, and evaluation results documented. The aim is that another student can understand how the result was produced and repeat the core experiment without relying on undocumented manual steps.

Planned Mitigation

I will begin with a small baseline and a feasibility check. If the original scope becomes too large for the available time or resources, I will narrow the research question rather than adding unnecessary complexity. Limitations and failed experiments will be documented as part of the research process.

Expected Outcomes

- A working research-workflow web prototype.
- A documented database and workflow design.
- Usability findings from realistic student scenarios.
- Recommendations for improving research collaboration tools.

Potential Research Collaboration

I am looking for someone interested in software engineering, UI/UX, databases, or research workflow design. A collaborator who enjoys testing interfaces and discussing practical student needs would be especially helpful.

Technical Stack

HTML, CSS, JavaScript, React or a lightweight frontend, Node.js/Express, MySQL, Git.

Scope, Ethics, and Responsible Research

The prototype will use demo or consent-based data. No sensitive student information is required. Access controls will be designed to prevent users from viewing project information that is outside their intended collaboration scope.

Proposed Work Plan

Phase	Activities
1. Background study	Literature review, problem definition, feasibility check
2. Setup	Data/system preparation and baseline implementation
3. Experiments	Development, comparison, tuning, and documentation
4. Evaluation	Metrics, error analysis, limitations, reproducibility
5. Reporting	Results, discussion, documentation, and presentation

What I Am Looking for in a Collaborator

I am looking for someone who is genuinely interested in the research question and willing to contribute to discussion, experiment design, implementation, analysis, or writing. I would like to divide responsibilities according to each person's strengths and keep the work transparent and documented.

Possible Next Step

If a researcher is interested, I would first like to discuss feasibility, refine the research question, and agree on a realistic division of work. The proposal can then be updated into a more specific study plan with milestones.

Note: This is a proposed student project. It does not claim completed research or guaranteed results, and the scope can be refined through collaboration.